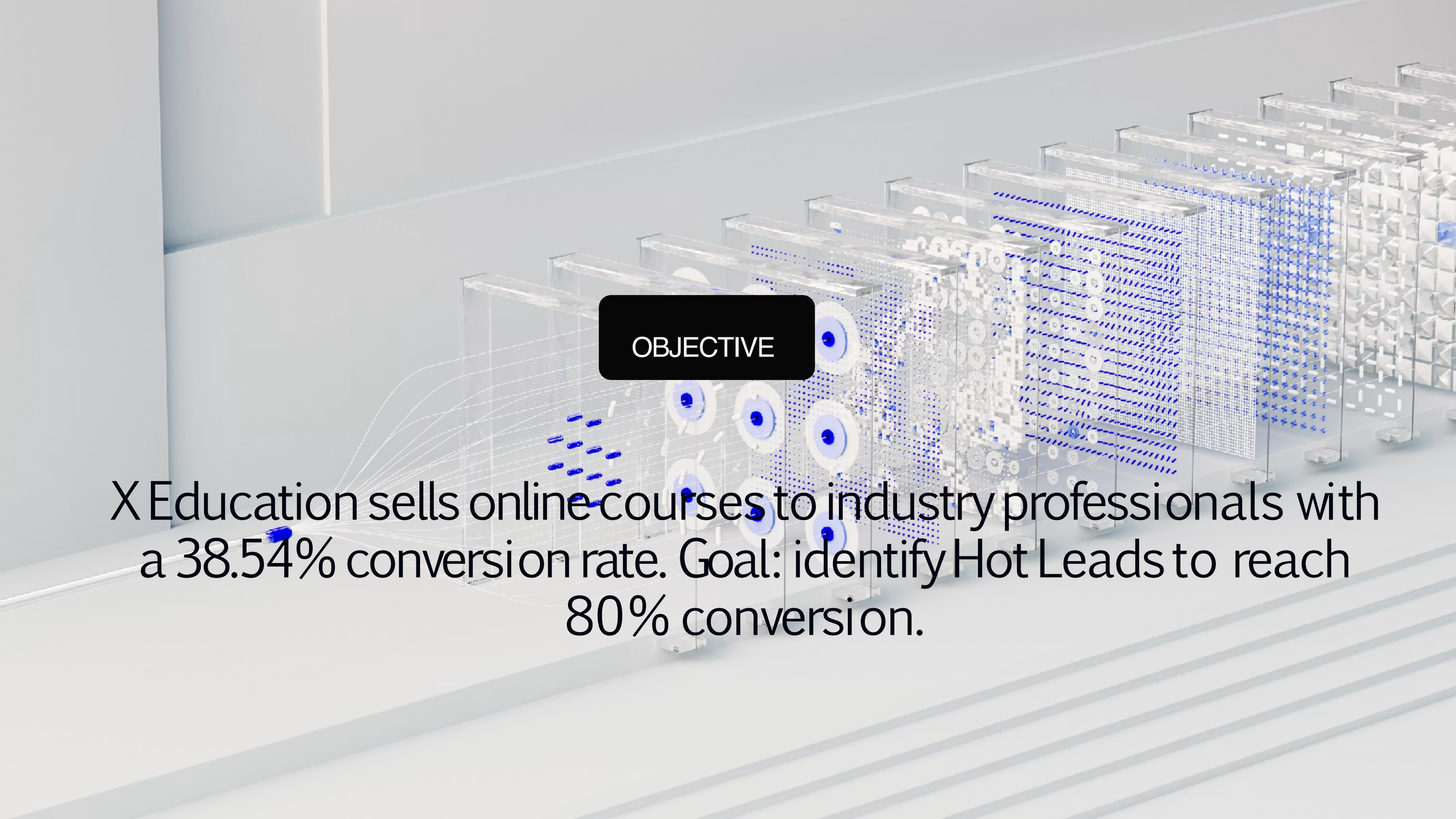
Lead Scoring Case Study

X EDUCATION | LOGISTIC REGRESSION MODEL FOR HOT LEAD IDENTIFICATION



Submitted by: Sakshi Ghodeswar



OBJECTIVE

X Education sells online courses to industry professionals with a 38.54% conversion rate. Goal: identify Hot Leads to reach 80% conversion.

Dataset Overview

TOTAL RECORDS

9,240

LEADS IN DATASET

VARIABLES

37

ORIGINAL COLUMNS

CONVERTED

3,561

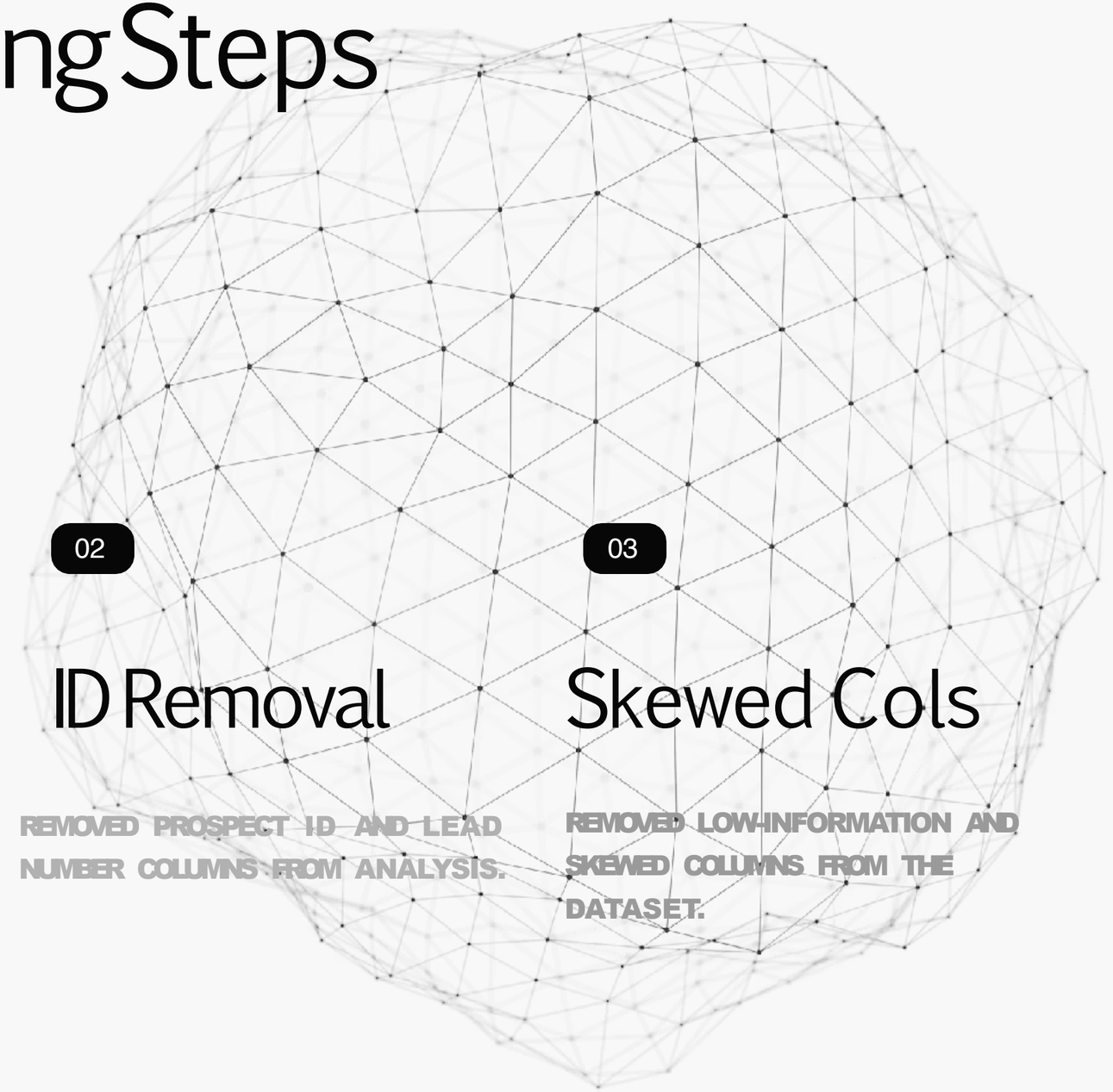
LEADS CONVERTED

CONVERSION RATE

38.54%

INITIAL BASELINE

Data Cleaning Steps



01

Missing Data

REPLACED "SELECT" WITH MISSING VALUES AND DROPPED HIGH-MISSING COLUMNS.

02

ID Removal

REMOVED PROSPECT ID AND LEAD NUMBER COLUMNS FROM ANALYSIS.

03

Skewed Cols

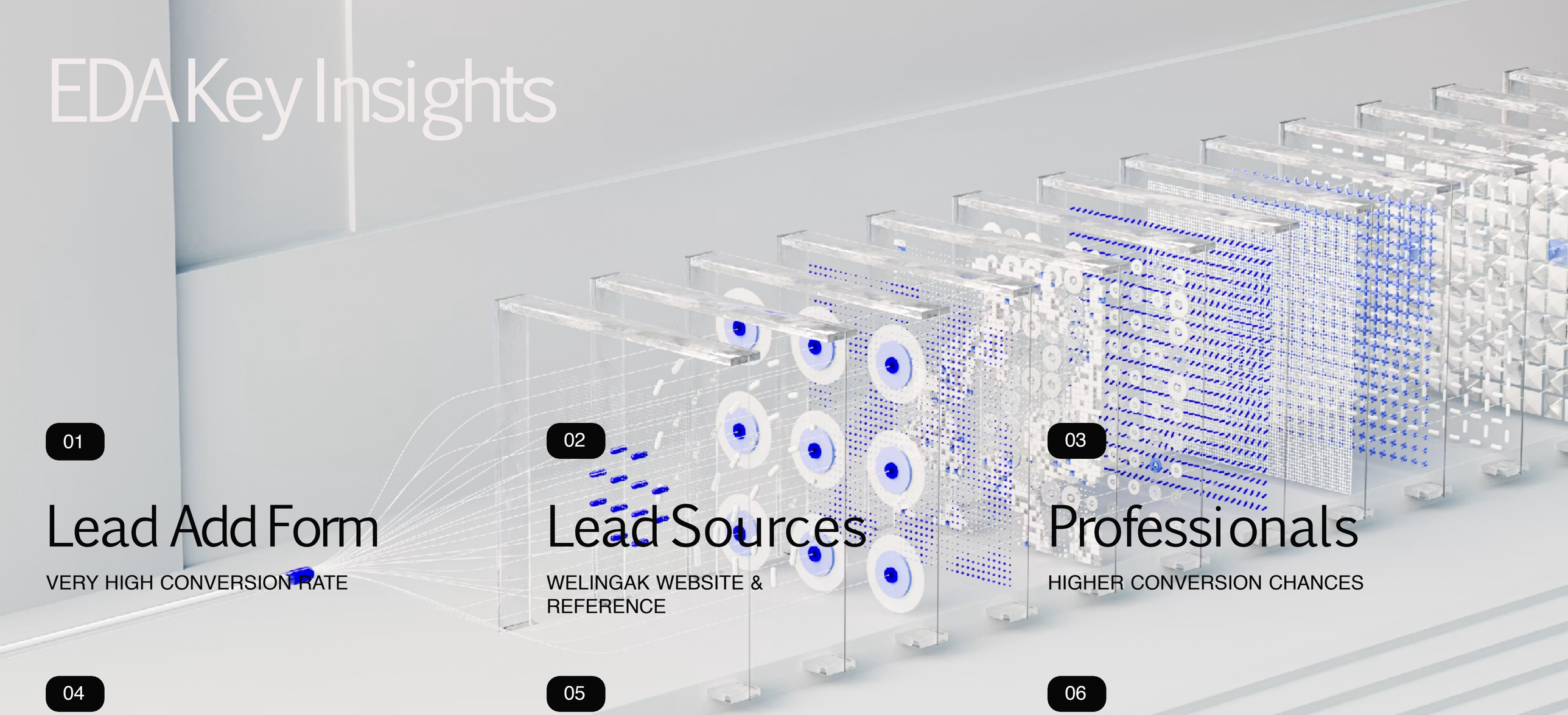
REMOVED LOW-INFORMATION AND SKEWED COLUMNS FROM THE DATASET.

04

Final Output

9,240 ROWS × 10 COLUMNS AFTER CLEANING, BEFORE DUMMY CREATION.

EDA Key Insights



01

Lead Add Form

VERY HIGH CONVERSION RATE

02

Lead Sources

WELINGAK WEBSITE & REFERENCE

03

Professionals

HIGHER CONVERSION CHANCES

04

Time on Site

CONVERTED SPENT MORE TIME

05

SMS Sent

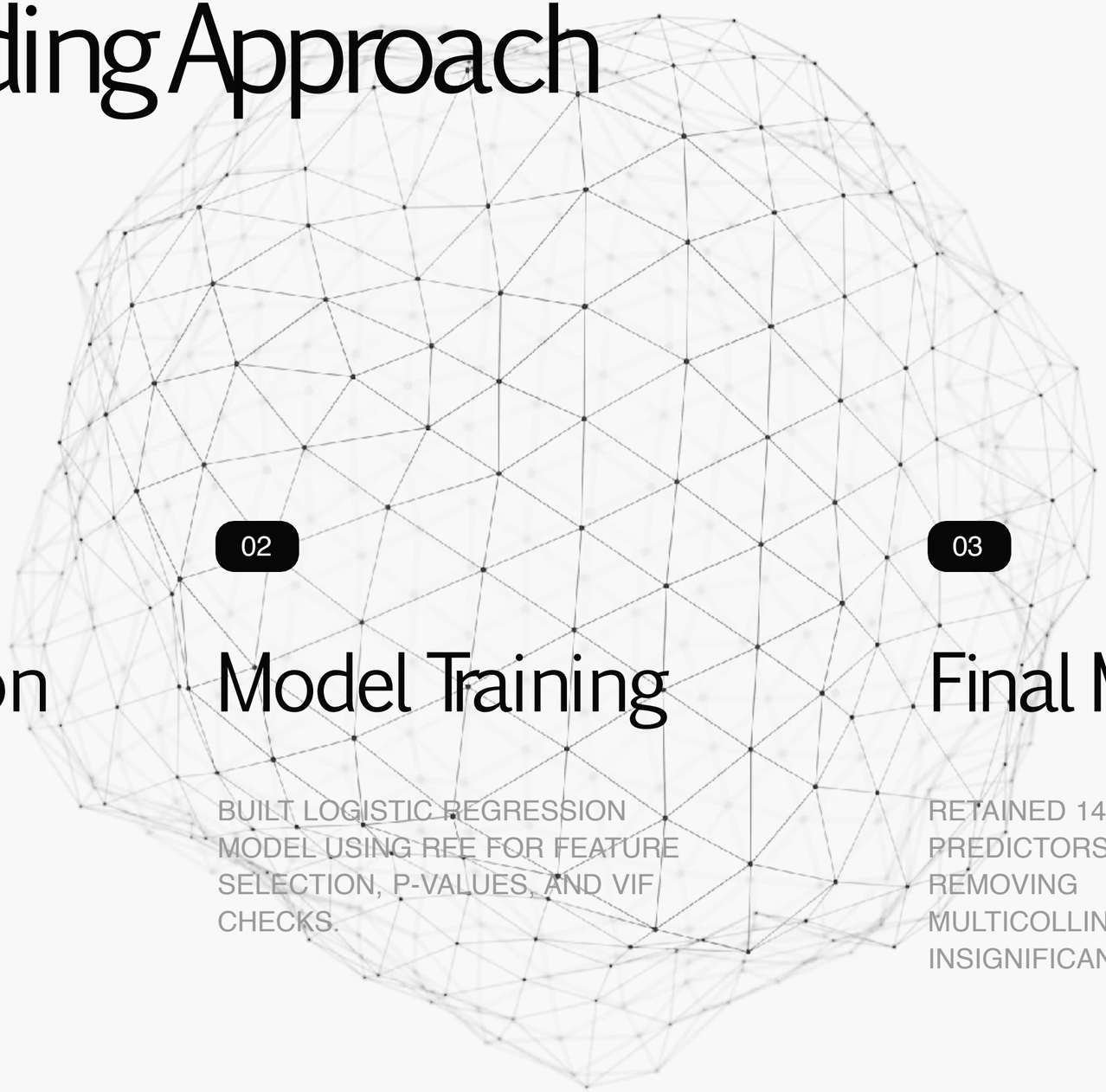
LINKED WITH BETTER CONVERSION

06

Do Not Email

"YES" HAD LOWER CONVERSION

Model Building Approach



01

Data Preparation

CONVERTED CATEGORICAL VARIABLES TO DUMMIES. SPLIT 70/30 TRAIN-TEST. SCALED WITH STANDARDSCALER.

02

Model Training

BUILT LOGISTIC REGRESSION MODEL USING RFE FOR FEATURE SELECTION, P-VALUES, AND VIF CHECKS.

03

Final Model

RETAINED 14 SIGNIFICANT PREDICTORS AFTER REMOVING MULTICOLLINEARITY AND INSIGNIFICANT FEATURES.

Final Model Performance

01

80.1%

TRAINING ACCURACY (SENS: 83.8%, SPEC: 77.7%)

02

80.2%

TEST ACCURACY (SENS: 83.7%, SPEC: 77.9%)

03

0.881

TRAIN AUC

04

0.883

TEST AUC OPTIMAL CUTOFF: 0.3

Important Variables

TOP POSITIVE PREDICTORS IDENTIFIED BY THE LOGISTIC REGRESSION MODEL



LEAD ADD FORM

FORM-BASED LEADS INDICATE HIGH PURCHASE INTENT AND COMMITMENT.

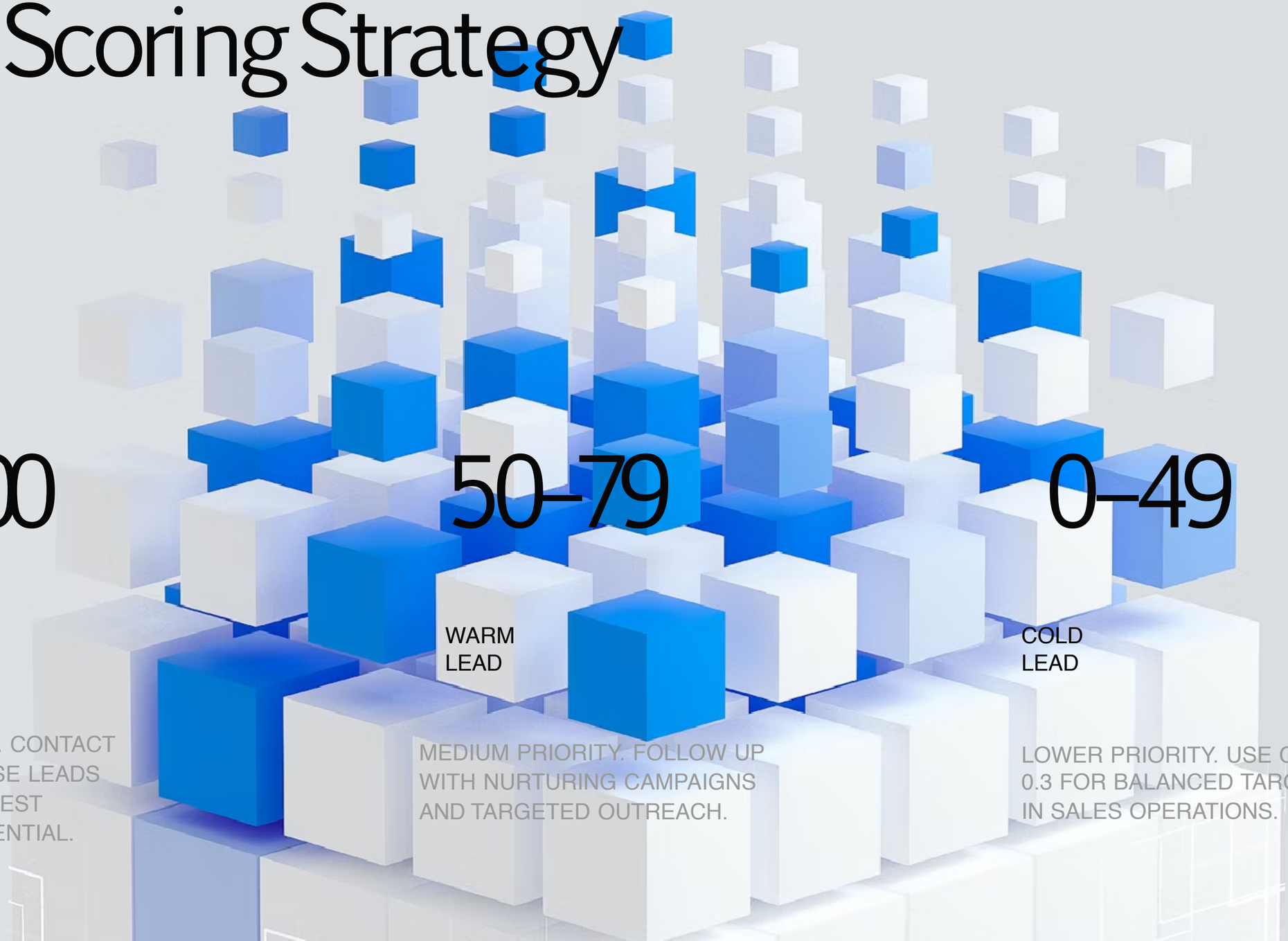
WORKING PROFESSIONAL

EMPLOYED PROFESSIONALS ARE MORE LIKELY TO INVEST IN CAREER COURSES.

WELINGAK WEBSITE

THIS LEAD SOURCE GENERATES THE HIGHEST QUALITY LEADS FOR CONVERSION.

Lead Scoring Strategy



80–100

HOT LEAD

HIGHEST PRIORITY. CONTACT IMMEDIATELY. THESE LEADS HAVE THE STRONGEST CONVERSION POTENTIAL.

50–79

WARM LEAD

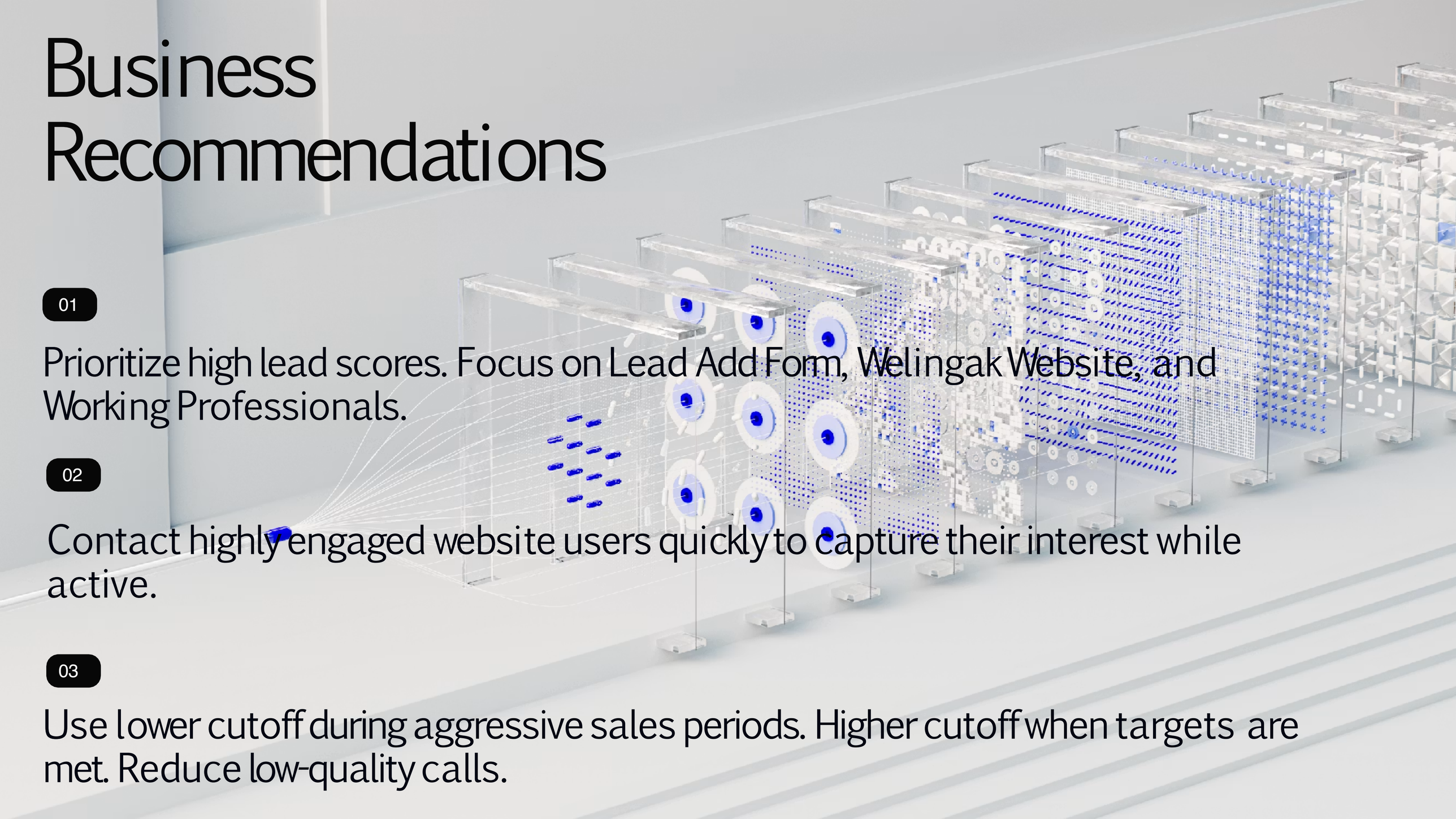
MEDIUM PRIORITY. FOLLOW UP WITH NURTURING CAMPAIGNS AND TARGETED OUTREACH.

0–49

COLD LEAD

LOWER PRIORITY. USE CUTOFF 0.3 FOR BALANCED TARGETING IN SALES OPERATIONS.

Business Recommendations



01

Prioritize high lead scores. Focus on Lead Add Form, Welingak Website, and Working Professionals.

02

Contact highly engaged website users quickly to capture their interest while active.

03

Use lower cutoff during aggressive sales periods. Higher cutoff when targets are met. Reduce low-quality calls.